Probability and Statistics (MT-2005)

Homework 04 (a) Statistical Properties of Discrete Probability Distributions

Negative Binomial Distribution

Question#1: A Negative Binomial distribution

$$X \sim \mathrm{NB}(r, p)$$

has the PMF:

$$P(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \quad k = r, r+1, r+2, \ldots$$

where r is the number of successes and p is the probability of success in each trial.

1. Derive the mean and variance of the Negative Binomial distribution.
2. Compute the first three moments about the origin.
3. Find the Moment Generating Function (MGF) and use it to derive the mean.
4. Compute the first two cumulants.
5. Derive the skewness and kurtosis of the Negative Binomial distribution.

Hypergeometric Distribution

Question#2:A Hypergeometric distribution

$$X \sim \mathrm{Hyper}(N, K, n)$$

has the PMF:

$$P(X = x) = \frac{\binom{k}{x}\binom{N-k}{n-x}}{\binom{N}{n}}, \quad \max(0, n-(N-k)) \leq k \leq \min(n, k)$$

where N is the population size, k is the number of success items in the population, and n is the sample size.

1. Derive the mean and variance of the Hypergeometric distribution.
2. Compute the first three moments about the origin.
3. Explain why Moment Generating Function (MGF) is not commonly used for the Hypergeometric distribution.
4. Compute the first two cumulants.
5. Derive the skewness and kurtosis of the Hypergeometric distribution.